METRIC
MIL-PRF-32684
26 August 2022

PERFORMANCE SPECIFICATION

VAPOR DEGREASING SOLVENT (VDS)

1. SCOPE

1.1 Scope. This performance specification covers the requirements for three types of vapor degreasing solvents. These solvents are suitable for use in degreasing applications for all military ground and aviation equipment wherein the cleaning processes employed is normally vapor phase contact with the metal parts to be cleaned but can also include a combination of submersion in an ultrasonically-agitated sump or boil sump or use of a spray wand or spray nozzles. These solvents are not intended to be used for painted parts or parts with rubber, plastic, or other non-metallic components. Further, as these solvents are at least partially halogenated, they should be considered similar to type 2A solvents as listed in ADS-61A-PRF.

1.2 Classification. The solvent is of the following three types:

Type	Description	Military Symbol
I	High VOC* solvent	VDS-1
II	Low VOC solvent	VDS-2
III	Very Low VOC solvent	VDS-3

*Volatile Organic Compounds

1.2.1 Type I. Vapor degreasing solvents with high Volatile Organic Compound (VOC) content (VOC greater than 50 g/L).

1.2.2 Type II. Vapor degreasing solvents with low VOC content (VOC greater than 25 g/L, less than or equal to 50 g/L).

1.2.3 Type II. Vapor degreasing solvents with very low VOC content (VOC less than or equal to 25 g/L).

Comments, suggestions, or questions on this document should be addressed to U.S. Army Ground Vehicle Systems Center, ATTN: FCDD-GVS-SAT MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000, or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6850

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-791	-	Testing Method of Lubricants, Liquid Fuels, and Related Products
MIL-PRF-81322	-	Grease, Aircraft, General Purpose, Wide Temperature Range
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base
O-M-232	-	Methanol (Methyl Alcohol)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ENVIRONMENTAL PROTECTION AGENCY STANDARDS

Method 3585	-	Waste Dilution for Volatile Organics
Method 8260	-	Volatile Organic Compounds by Gas Chromatography/Mass Spectrometry (GC/MS)
Method 9065	-	Phenolics, (Spectrophotometric, Manual 4-AAP with Distillation)

(Copies of these documents are available at https://www.epa.gov.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

METHOD 313	-	Determination of Volatile Organic Compounds (VOC) by Gas Chromatography/Mass Spectrometry/Flame ionization Detection (GC/MS/FID)

(Copies of this document are available at http://www.aqmd.gov/home/regulations/compliance/vocs/working-group.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from https://asq.org.)

ASTM INTERNATIONAL

ASTM D56	-	Standard Test Method for Flash Point by Tag Closed Cup Tester
ASTM D86	-	Standard Test Method for Distillation of Petroleum Products and Liquid Fuels at Atmospheric Pressure
ASTM D130	-	Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test
ASTM D740	-	Standard Specification for Methyl Ethyl Ketone
ASTM D2109	-	Standard Test Methods for Nonvolatile Matter in Halogenated Organic Solvents and Their Admixtures
ASTM D2879	-	Standard Test Method for Vapor Pressure-Temperature Relationship and Initial Decomposition Temperature of Liquids by Isoteniscope
ASTM D3244	-	Standard Practice for Utilization of Test Data to Determine Conformance with Specifications
ASTM D3257	-	Standard Test Methods for Aromatics in Mineral Spirits by Gas Chromatography
ASTM D4057	-	Standard Practice for Manual Sampling of Petroleum and Petroleum Products

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

ASTM D4080	-	Standard Specification for Trichloroethylene, Technical and Vapor-Degreasing Grade
ASTM D4177	-	Standard Practice for Automatic Sampling of Petroleum and Petroleum Products
ASTM D4376	-	Standard Specification for Vapor-Degreasing Grade Perchloroethylene
ASTM E29	-	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications
ASTM E1210	-	Standard Test Method for Fluorescent Liquid Penetrant Examination Using the Hydrophilic Post-Emulsification Process
ASTM E1219	-	Standard Test Method for Fluorescent Liquid Penetrant
ASTM E1417/E1417M	-	Standard Practice for Liquid Penetrant Inspection
ASTM F483	-	Standard Practice for Total Immersion Corrosion Test for Aircraft Maintenance Chemicals
ASTM F945	-	Standard Test Method for Stress-Corrosion of Titanium Alloys by Aircraft Engine Cleaning Materials
ASTM F1110	-	Standard Test Method for Sandwich Corrosion Test
ASTM F1111	-	Standard Test Method for Corrosion of Low-Embrittling Cadmium Plate by Aircraft Maintenance Chemicals

(Copies of these documents are available from https://www.astm.org.)

SAE INTERNATIONAL

AMS-M-3171	-	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on
AMS-QQ-A-250	-	Aluminum and Aluminum Alloy, Plate and Sheet, General Specification for
AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)
AMS 2470	-	Anodic Treatment of Aluminum Alloys Chrome Acid Process
AMS 2644	-	Inspection Materials, Penetrants
AMS 4037	-	Aluminum Alloy, Sheet and Plate 4.4Cu - 1.5Mg - 0.60Mn (2024- -T3

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

		Flat Sheet, -T351 Plate) Solution Heat Treated
AMS 4041	-	Aluminum Alloy, Alclad Sheet Plate 4.4Cu - 1.5Mg - 0.60Mn Alclad 2024 and 1-1/2% Alclad 2024, -T3 Flat Sheet- 1-1/2% Alclad 2024-T351 Plate
AMS 4045	-	Aluminum Alloy Sheet and Plate, 5.6Zn - 2.5Mg - 1.6Cu - 0.23Cr 7075: (-T6 Sheet, -T651 Plate), Solution and Precipitation Heat Treated
AMS 4049	-	Aluminum Alloy, Sheet and Plate, Alclad, 5.6Zn - 2.5Mg - 1.6Cu - 0.23Cr (Alclad 7075; -T6 Sheet - T651 Plate) Solution and Precipitation Heat Treated
AMS 4377	-	Magnesium Alloy, Sheet and Plate, 3.0Al - 1.0Zn - 0.20Mn (AZ31B-H24), Cold Rolled, Partially Annealed
AMS 4904	-	Titanium Alloy, Sheet, Strip, and Plate, 6Al - 4V Solution Heat Treated and Aged
AMS 5549	-	Steel, Corrosion and Heat Resistant, Plate, 15.5Cr - 4.5Ni - 2.9Mo - 0.10N, Solution Heat Treated
AMS 5629	-	Steel, Corrosion-Resistant, Bars, Wire, Forgings, Rings, and Extrusions 13Cr - 8.0Ni - 2.2Mo - 1.1Al Vacuum Induction Plus Consumable Electrode Melted Solution Heat Treated, Precipitation Hardened (H1000)
AMS 6345	-	Steel, Sheet, Strip, and Plate 0.95Cr - 0.20Mo (0.28 - 0.33C) (SAE 4130) Normalized or Otherwise Heat Treated
AMS 6414	-	Steel, Bars, Forgings, and Tubing 0.80Cr - 1.8Ni - 0.25Mo (0.38 - 0.43C) (SAE 4340) Vacuum Consumable Electrode Remelted
AMS 6512	-	Steel, Maraging Bars, Forgings, Tubing, and Rings 18Ni – 7.8Co – 4.9Mo – 0.40Ti – 0.10Al Consumable Electrode Vacuum Melted, Annealed

(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence unless otherwise

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

noted herein or in the contract. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Solvents furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3). Any change to the formulation of a qualified product will necessitate its requalification.

3.2 Materials. The solvent shall be a virgin grade or recycled azeotrope solvent containing various fluorinated solvents and trans-dichloroethylene products derived from petroleum distillates, fraction from reclaiming and re-refining processes, or a mixture of these fractions. The resultant solvent must be produced in such a manner as is necessary to meet the specified requirements.

3.3 Recycled, recovered, environmentally preferable, or bio-based materials. Recycled, recovered, environmentally preferable, or bio-based materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Appearance. The solvent shall be clear and free from suspended matter and undissolved water when observed at a temperature between 15.6 and 25.6 °C (60.1 and 78.1 °F).

3.5 Toxicity. The solvent shall have no adverse effects on human health when used as intended (see 6.1 and 6.6).

3.6 Carcinogenicity. The solvent shall contain no chemicals listed as carcinogens (see 6.4). Any carcinogenic components in the solvent in a concentration of 0.1 percent (%) or greater by weight (wt) or volume (vol) will be regarded as the presence of a carcinogen in the solvent.

3.7 Hazardous air pollutants. The solvent shall contain no chemicals listed as Hazardous Air Pollutants (HAPs) (see 6.4.2). Any HAP components in the solvent in a concentration of 1.0 percent (%) or greater by weight (wt) or volume (vol) will be regarded as the presence of a HAP in the solvent. The product containing less than 1.0 percent (%) HAP shall be considered as a HAP-free solvent. For carcinogenic HAPs (see 3.6).

3.8 Physical and chemical property requirements. The types listed in this specification shall comply with the chemical and physical property requirements listed in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

TABLE I. Physical and chemical property limits.

Property	Method	Type I	Type II	Type III
Flash Point, °C	ASTM D56 (Manual)*	No flash	No flash	No flash
Equilibrium Boiling Point, °C, min	ASTM D86	40	40	40
Non-Volatile Residue, mg/100 mL, max	ASTM D2109 Method D	2.5	2.5	2.5
Volatile Organic Compounds (VOC), g/L	SCAQMD Method 313	VOC > 50	50 ≥ VOC > 25	VOC ≤ 25
Vapor Pressure at 20 °C, mmHg, max	ASTM D2879	300	300	300
Copper Corrosion, 3 h at 40 °C, rating, max	ASTM D130	1b	1b	1b
Aromatic Content, vol %, max	ASTM D3257	1	1	1
Total Phenol Content, ppm, max	EPA Method 9065	0.5	0.5	0.5
Benzene, mg/L, max	EPA Methods 3585 & 8260B	0.5	0.5	0.5
Dichlorobenzene, mg/L, max	EPA Methods 3585 & 8260B	0.5	0.5	0.5
Tetrachlorobenzene, mg/L, max	EPA Methods 3585 & 8260B	0.7	0.7	0.7
Trichloroethylene, mg/L, max	EPA Methods 3585 & 8260B	0.5	0.5	0.5

*ASTM D56 flash point testing of vapor degreasing solvents can display an enlarged test flame and/or halo. This does not constitute a flash point and should be ignored.

3.9 Materials compatibility. All solvents shall be compatible with the material testing listed in sections 3.9.1, 3.9.2, 3.9.3, 3.9.4 and 3.9.5.

3.9.1 Total immersion corrosion. There shall be no staining, etching, pitting, or weight change exceeding limits noted in table II. Testing shall be performed in accordance with ASTM F483.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

TABLE II. Alloy and maximum average weight loss.

Alloy (Test Panel)	Average* weight loss, maximum (mg/cm^2/168 h)
Stainless steel, AMS 5549, AM-355 CRT (for 14 mil material)	0.49
Stainless steel, AMS 5629, H1000 condition	0.49
Maraging 250 steel, AMS 6512	0.49
Magnesium alloy, AMS 4377, surface treated in accordance with AMS-M-3171 Type III	0.70
Titanium alloy, AMS 4904, 6AL-4V	0.35
Aluminum alloy, AMS 4037	0.49
Aluminum alloy, AMS 4041, Alclad	0.49
Aluminum alloy, AMS 4045	0.49
Aluminum alloy, AMS 4049, Alclad	0.49
Aluminum alloy, AMS-QQ-A-250, T3 surface treatment in accordance with AMS 2470	0.35
Steel alloy, AMS 6414, Oil quench 800°C/1425°F, 595°C /1100°F temper	0.49

*Average for three test panels.

3.9.2 Titanium stress corrosion. There shall be no cracks in titanium alloy (6Al-4V). Testing shall be performed in accordance with ASTM F945 with the following exceptions: Exclude testing of three restrained test specimens in a 3wt% sodium chloride solution. That is, test with a minimum of six specimens instead of nine. Three specimens remain untreated after acid cleaning, and the other three specimens are tested in the candidate solvent. (2) Replace furnace heating Method A (482±10°C (900±20°F) for 8±0.2 hours) with the following: Place three test specimens in a 500 ml reaction flask fitted with multiple ports and closed with an O-ring seal. Add sufficient solvent to form a layer of solvent one inch deep. Attach a reflux condenser, insert a thermometer or thermocouple, and heat over a light shielded 150-watt light bulb and reflux for 8 hours a day for 5 days. Record the temperature of the heated solvent during the testing. The bulb heating can be replaced by heating with a bath type apparatus where the bath temperature is controlled.

3.9.3 Sandwich corrosion. There shall be a rating of 1 when tested with SAE AMS 4037, AMS 4041, AMS 4045 and AMS 4049. Testing shall be performed in accordance with ASTM F1110.

3.9.4 Low-embrittling cadmium plate corrosion. There shall be no weight change greater than 0.14 mg/cm^2/24 hrs. Testing shall be performed in accordance with ASTM F1111. Test specimens shall be made of 4130 steel Condition N in accordance with AMS 6345,

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

measuring 25.4 by 50.8 by 1.22 mm and cadmium plated in accordance with AMS-QQ-P-416, Type I.

3.9.5 Fluorescent penetrant inspection. The following procedure shall be conducted for the fluorescent penetrant inspection. The entire procedure shall be repeated three times. The candidate vapor degreasing solvent must permit all cracks to be identified and the brightness intensity must be greater than or equal to the control cleaner.

a. Prepare the 25 x 150 x 6 mm (1 x 6 x 0.25 inch) test specimen bars with the fatigue-induced cracks according to table III.

TABLE III. Fatigue-induced crack length.

Alloy (Test Panel)	Crack Length
Inconel 718	0.50 ± 0.05 mm (0.020 ± 0.002 inch)
Inconel 718	1.50 ± 0.25 mm (0.060 ± 0.010 inch)
Ti-6Al-4V	1.50 ± 0.25 mm (0.060 ± 0.010 inch)
Ti-6Al-4V	9.50 ± 0.50 mm (0.375 ± 0.020 inch)

b. Pre-clean the test specimen bars as follows: Hand wash with soapy water, e.g., Alconox solution. Rinse in hot tap water and wipe dry with clean cloth. Ultrasonically clean in methanol (0-M-232 Grade A) for 15±0.5 minutes. Ultrasonically clean in perchloroethylene (ASTM D4376), trichloroethylene (ASTM D4080), or methyl ethyl ketone (ASTM D740) for 30±0.5 minutes. Check the bars under ultraviolet (black) light to ensure that all penetrant has been removed from the bar surfaces/cracks. If penetrant remains, repeat above steps. Dry bars in oven at 55±8.3°C (130±15°F) for five minutes and allow to cool to room temperature.

c. The standard contaminant shall be prepared by mixing two parts by weight of hydraulic fluid (MIL-PRF-83282 or equivalent), one part by weight lubricating grease (MIL-PRF-81322 or equivalent), and one-tenth by weight of carbon black.

d. Apply the standardized contaminant mixture to the test bars as follows: Using a clean glass rod apply two drops of the contaminant to each bar, and evenly spread the contaminant over the surface of each bar in the area of the crack. Place the bars in a dry air-circulating oven for a minimum of two hours at 55±8.3°C (130±15°F).

e. Wipe excess contaminant from the bars with a clean lint-free cloth. Assemble a reflux apparatus with the candidate cleaner following steps A.7.4 through A.7.9 of the Appendix. Place the contaminated bars into the reaction kettle for 10±0.5 minutes.

f. Place the bars in an oven at 55±8.3°C (130±15°F) for 5±0.5 minutes. Remove the bars from the oven and allow them to cool at room temperature.

g. Penetrant inspect in accordance with ASTM E1417/E1417M or ASTM E1210, using penetrant materials that are qualified per AMS 2644. If classification is not known, use Type I (fluorescent dye), Method D Process (post emulsifiable, hydrophilic), and Level 3 or Level 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

(high or ultrahigh) sensitivity penetrant, and Form A (dry powder) developer. Record the product names and classification of penetrant, emulsifier, and developer used. If Method C (solvent removable) method is being tested with the replacement candidate solvent, apply 1-2 drops of Level 3 or 4 penetrant via glass rod to the cracked area on the precleaned test bars. Allow a minimum of 10 minutes for the penetrant to dwell, and then remove all excess penetrant via wiping once with a solvent dampened (not soaked) lint free cloth. Over-removal of the surface penetrant shall require the test bars to be cleaned and re-processed.

h. Measure the fluorescent brightness intensity of the penetrant as follows: Calibrate, zero, and focus a suitable photometer/spotmeter (e.g., Photo Research Photomultiplier-Tube Optical Photometer {Model UBD, PR-1500 Spectra Spotmeter}). Illuminate the cracked bar area of interest with high intensity UV source at greater than 10,000 $\mu W/cm^2$ verified via appropriate UV meter. Place specimen bars under ultraviolet (black) light per ASTM E1417/E1417M, ASTM E1219 or ASTM E1210. For each bar, measure the intensity value of the crack indication and the background intensity value on the bar surface near the crack WITHOUT MOVING THE LIGHT SOURCE OR THE PHOTOMETER. Subtract the background reading value from the crack reading value to obtain the relative intensity of the crack indication. Record all measurements and observations. Take all measurements in similar sequence and minimize the amount of time between readings to ensure similar dwell times of the penetrant.

i. Repeat steps b through h but replace the candidate cleaner/process in step e with the control cleaning method (e.g., cleaning with methyl ethyl ketone {ASTM D740}, or vapor degreasing with perchloroethylene {ASTM D4376} or trichloroethylene {ASTM D4080}).

j. Compare the brightness intensities obtained with the candidate and the control cleaners.

k. When all testing has been completed, clean specimen bars as described in step b to remove all residual penetrant inspection materials.

3.10 Cleaning Performance. The types listed in this specification shall comply with the cleaning performance requirements listed in table IV.

TABLE IV. Cleaning performance requirements.

Property	Method	Type I	Type II
Ultrasonic Solvency, %, min	FED-STD-791, Method 7502	90	90
Vapor Phase Solvency, %, min	See Appendix A	90	90

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspections (see 4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

b. Conformance inspections (see 4.3).

4.1.1 Inspections. Unless otherwise specified, all inspections shall be performed in accordance the test conditions specified in 4.4.

4.2 Qualification inspections. Qualification inspections shall consist of all verifications listed in Section 3.

4.3 Conformance inspections. Conformance inspections shall consist of a sample for conformance verification purposes (see 4.3.2), samples for examination from filled containers (see 4.3.3), and the verifications specified in table I.

4.3.1 Lot. A lot shall consist of solvents from one batch or tank offered for delivery at one time. If material cannot be identified by batch or lot, a lot shall consist of not more than 10,000 gallons offered for delivery at one time (see 6.2).

4.3.2 Sampling for tests. Sampling of a lot for conformance inspection purposes shall be in accordance with ASTM D4057 or D4177.

4.3.3 Sample for examination of filled containers. A random sample from filled unit containers and a sample of shipping containers fully prepared for delivery shall be selected from each lot of solvent in accordance with ASQ/ANSI Z1.4.

4.4 Verification test procedures. Tests shall be conducted as specified in table I. For purposes of determining conformance with each requirement, an observed value or calculated value shall be rounded off to the nearest unit in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off procedure given in ASTM E29. All values, unless otherwise stated or agreed upon by the qualifying activity are critical specifications as defined by ASTM D3244.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in- house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

6.1 Intended use. The solvent is a virgin grade, a recycled azeotrope solvent or a mixture thereof containing a trans-dichloroethylene product fraction and is used as a degreaser and cleaner for unpainted metal parts. These solvents are not intended to be used for painted parts or parts with rubber, plastic, or other non-metallic components. It is recommended that these solvents should be used with an appropriate vapor degreasing apparatus. Cleaning processes could include only vapor phase cleaning or a combination of submersion into ultrasonically-agitated sump, submersion into boil sump, spray wand, spray nozzles and vapor phase cleaning. No other commercial specification is available to cover these military applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type and quantity (see 1.2 and 4.3.1).

c. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List (QPL) No. 32684 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from:

U.S. Army Futures Command
Combat Capabilities Development Command (DEVCOM)
Ground Vehicle Systems Center (GVSC)
Force Projection Technology (FPT)
ATTN: Fuels & Lubricants Branch
FCDD-GVS-ES-FPT (MS-110)
6501 E. 11 Mile Road
Warren, MI 48397-5000
usarmy.detroit.devcom-gvsc.mbx.pol-help@army.mil

An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Definitions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

6.4.1 Carcinogens. The Occupational Safety and Health Administration (OSHA) definition of carcinogens are those chemicals / processes appearing in lists 1, 2A, and 2B of the International Agency for Research on Cancer (IARC) [1/]; substances known to be carcinogenic and occupational exposures associated with a technological process known to be carcinogenic by the National Toxicology Program (NTP) Report on Carcinogens (latest annual report) [2/]; and OSHA regulated carcinogens.

6.4.2 Hazardous Air Pollutants (HAP). HAP is defined as any substance listed under Section 112 of the Clean Air Act or its modifications. The text of the Clean Air Act, listed pollutants and modifications are kept by the Environmental Protection Agency (EPA) and are accessible through the website: https://www.epa.gov

NOTE:
1. Copies of these volumes may be found in medical libraries or through the World Health Organization, 1211 Geneva 27, Switzerland or at https://www.who.int

2. Copies may be obtained from the Public Health Service, National Toxicology Program, Public Information Office, P.O. Box 12233, MD B2-04, Research Triangle Park, NC 27709 or at https://www.niehs.nih.gov

6.5 Disposal actions. Disposal of this product should be in accordance with local, state and Federal regulations. Care should be taken to avoid mixing used MIL-PRF-32684 with other waste materials, as they can contain unsaturated halogenated hydrocarbons.

6.6 Safety data sheets. Contracting officers will identify those activities requiring copies of completed Safety Data Sheets (SDSs) prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313; and 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the SDS (see 3.5).

6.7 Solvent fit for purpose monitoring. During vapor degreasing operation, it is recommended for the user to coordinate with the solvent manufacturer in developing a process to best monitor and maintain the solvent chemistry to ensure the solvent remains fit for purpose. While the process and frequency are dependent on the solvent type, vapor degreasing system, and operational environment, it is recommended for at least annual testing of the solvent. Extract a solvent sample to conduct a minimum of the following tests.

a. Flash Point ASTM D56 (Manual) conforming to requirement in table I.

b. Equilibrium Boiling Point ASTM D86 conforming to requirement in table I.

c. No liquid separation when visually inspected.

6.8 Subject term (key word) listing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

Solvent, Azeotrope
Solvent (Type II), Low VOC
Solvent (Type III), Very Low VOC

6.9 International standardization agreement implementation. Certain provisions of this specification are the subject of international standardization agreement (NATO STANAG/AFLP 1135 Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations). When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

VAPOR DEGREASING SOLVENT CLEANING POWER BY A SOIL TEST METHOD

A.1. SCOPE

A.1.1 Scope. This test method is used for determining the cleaning power of vapor degreasing solvents. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2. APPLICABLE DOCUMENT

SAE INTERNATIONAL

SAE J2360	-	Automotive Gear Lubricants for Commercial and Military Use

(Copies of these documents are available from https://www.sae.org.)

A.3. SUMMARY OF METHOD

A.3.1 An aluminum coupon is cleaned, weighed and dipped into 23±3°C SAE J2360 gear oil. The coupon is suspended for 60±2 minutes in 23±3°C air to allow for dripping of the oil. The coupon with an oil film is weighed before being immersed into the vapor of the boiling test solvent for 60±2 seconds. The oil is removed from the aluminum coupon by solvent vapor condensing on the aluminum coupon, the condensed vapor dissolving oil, and gathering of condensate gravitationally dripping back into the reservoir. The coupon is weighed after the vapor cleaning process. The solvent cleaning power is calculated based on the respective weights before and after the cleaning process.

A.4. SIGNIFICANCE AND USE

A.4.1 This test method measures vapor degreasing solvent cleaning power of the vapor phase.

A.5. APPARATUS

A.5.1 Distillation column.
A.5.2 Reaction Kettle, 1000mL.

A.5.3 Reaction Kettle Lid with at least three necks and clamp.

A.5.4 Hotplate.
A.5.5 Support stand and test tube clamps.
A.5.6 Water Circulation system with tubing.
A.5.7 Test specimen assembly. Aluminum coupon 1 cm wide, 5cm long, and 0.081 cm thick with holding wire (see figure A-1). 2024 aluminum, 500 ± 200 nm Ra surface roughness.
A.5.8 Timer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

A.5.9 RTD (resistance temperature detector) thermometer, 200°C with accuracy ≤ 0.5 °C.

A.5.10 Rubber Stoppers, one with center hole

A.5.11 Analytical balance with accuracy ≤ 0.1mg.

A.5.12 Two 200mL tall form beakers

A.6. MATERIALS

A.6.1 Hexane of certified purity ≥ 99% (Caution- Flammable, vapor harmful)

A.6.2 SAE J2360 75W-90 Gear Oil

A.7. PROCEDURE

A.7.1 Clean a 200mL tall form beaker with hexanes. Place the beaker on an analytical balance and tare the balance.

A.7.2 Holding the wire, wash the test specimen assembly with hexane until free from any soil or oil film. Dry the test specimen assembly using laboratory air system. Manipulate the wire to suspend the test specimen assembly in the beaker on the analytical balance. Record the weight to the nearest 0.1mg as value A.

A.7.3 Holding the wire, submerge the aluminum coupon into 23±3°C SAE J2360 gear oil for 5 seconds. Withdraw the aluminum coupon from the oil and suspend the assembly using the second 200mL tall form beaker, allowing excess oil to drip for 60±2 minutes. Suspend the assembly with the oil film in the beaker on the analytical balance. Record the weight to the nearest 0.1mg as value B.

A.7.4 Fill the 1000mL reaction kettle with 250±10mL of the vapor degreasing solvent to be tested. Using the reaction kettle clamp, affix the kettle lid onto the kettle. Set the kettle on a hotplate.

Note: All solvent work should be conducted in an operating fume hood.

A.7.5 Place a center-hole rubber stopper into one of the necks of the kettle lid, and then place an RTD thermometer through the center-hole of the stopper so that the tip of the probe is in the vapor zone.

A.7.6 Occupy the center kettle lid neck with a distillation column. Affix the distillation column with a support stand and test tube clamp. Cycle 23±3°C water through the jacket of the distillation column. Allow the top of the distillation column to be open to atmosphere to prevent pressure build-up.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

A.7.7 Place stoppers into the remaining necks of the kettle lid. The assembled vapor degreasing test apparatus should appear as pictured in figure A-2.

A.7.8 Turn on the hotplate and monitor the RTD thermometer temperature. Once the boiling point of the vapor degreasing solvent is reached, the temperature should stop rising and condensation droplets from the distillation column should be evident. Thus, the process of boiling, vaporization, condensing and recirculating solvent is achieved.

A.7.9 Once the vapor degreasing apparatus reaches thermal equilibrium at the boiling temperature of the solvent, record the temperature to the nearest 0.1°C. The test temperature should remain at the boiling temperature of the solvent ±3°C.

A.7.10 Place the test specimen assembly into one of the necks with a stopper in the kettle lid. Adjust the length of wire accordingly to position the test coupon into the vapor region of the kettle. Place the stopper back in with the wire to hold the assembly in place. Start the test using a timer.

A.7.11 Once the timer reaches 60±2 seconds, remove the test specimen assembly from the kettle. Suspend the assembly in the beaker on the analytical balance. Record the weight to the nearest 0.1mg as value C.

A.7.12 Repeat two more tests in accordance with A.7.1 through A.7.11.

A.8. CALCULATION

A.8.1 Calculate the cleaning power as follows:

$$Solvent\ cleaning\ power, \% = \frac{B - C}{B - A} * 100$$

Where:
A is the test specimen assembly weight recorded in A.7.2.
B is the test specimen assembly with oil weight recorded in A.7.3.
C is the test specimen assembly after cleaning weight recorded in A.7.11.

A.9. REPORT

A.9.1 Report the solvent cleaning power in percentage for all three trials and an average value to the nearest 0.1%.

A.10. PRECISION

A.10.1 Test precision has not yet been developed for the method.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

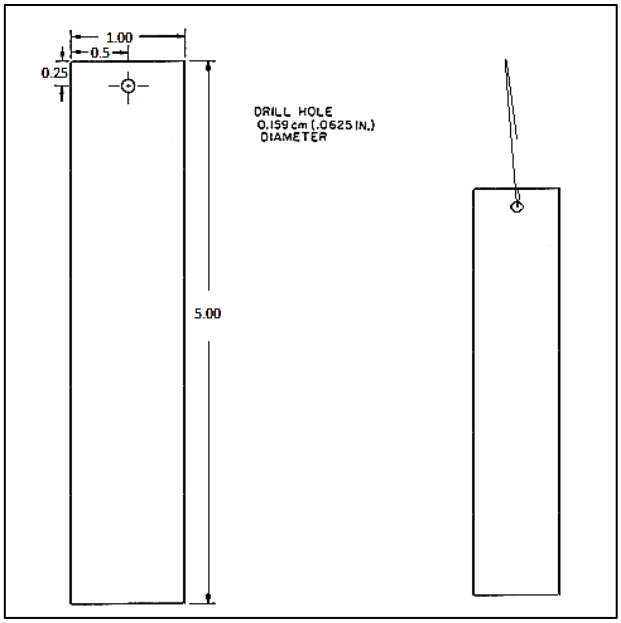


FIGURE A-1. Aluminum test coupon and test assembly.

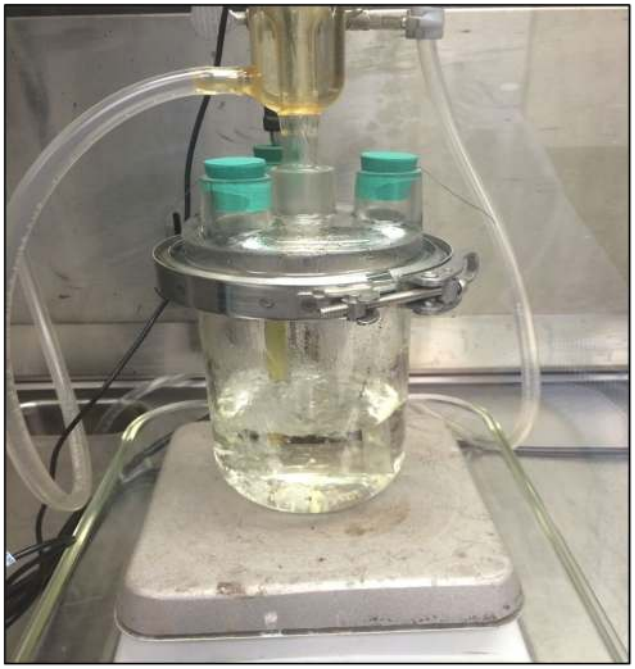


FIGURE A-2. Vapor degreasing test apparatus.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - AT
Air Force - 68
Navy – AS

Preparing Activity:
Army - AT

(Project 6850-2021-011)

Review Activities:
Army - AV, EA, MD1, MI, MR, SM
Air Force - 03, 20, 50
Navy - MC, SH
DLA - GS, GS3
DTRA – DS
GSA/GSS – FAS
MISC – MP

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:33Z
Check the source to verify that this is the current version before use.